

A-HEALTH

SYSTEM ANALYSIS & DESIGN — VERSION 2

Weighted, Blockchain-Anchored Architecture for African Health Systems

20-Week Implementation Plan

Prepared for the A-health Program
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1. Introduction

1.1 Purpose

This document defines the system analysis and design for A-health Version 2: a proactive healthcare platform for African health environments, built on a hospital-blockchain architecture. It translates the five weighted problem-solution pairs from the Version 2 solution brief into concrete functional requirements, a system architecture, a database design, an API surface, security controls, core use cases, and a 20-week build sequence.

1.2 Scope

The scope covers the patient-facing web and mobile applications, the USSD/SMS access channel for feature phones, the clinician and gig-doctor experience, the administrative verification workflow, and the shared blockchain layer that anchors identity, payments, clinical records, and consent across all five solution modules.

1.3 Guiding Priority Weights

Priority	Core Gap	Weight	Primary Solution
1	Facility Overburden	40%	Decentralized Telemedicine Network
2	Unemployed Doctors	20%	Gig Work Platform with Fair Payments
3	Follow-Up Void	15%	Daily Automated Digital Checkups
3	Medication Misuse	15%	Automated Adherence Reminders
5	Late Pathology Discovery	10%	AI Risk Screening + Early Detection

These weights drive both requirement priority (Section 2) and build order (Section 9): the two heaviest gaps, together 60% of impact, are engineered first.

2. Functional Requirements by Module

2.1 Telemedicine & Routing (Facility Overburden — 40%) *(Priority 1)*

Lets a patient open a consult from web, mobile, or USSD/SMS and be routed to the nearest available clinician without a physical queue.

- Patient can initiate a consult request via app, SMS, or USSD with symptom intake.
- Smart-routing engine matches request to nearest available licensed clinician by specialty, load, and geolocation (PostGIS).
- Real-time queue status visible to patient and clinician dashboards.
- Consult completion writes a hashed record to the clinical ledger.
- Emergency-transport request escalates to rapid-dispatch workflow when flagged urgent.

2.2 Gig-Doctor Marketplace (Unemployed Doctors — 20%) *(Priority 2)*

Lets verified doctors accept flexible shifts or consults and be paid automatically on completion.

- Doctor registration with credential upload and on-chain credential verification.
- Shift/consult marketplace showing available demand by region and specialty.
- Smart-contract escrow holds payment and releases automatically on verified service completion.
- Doctor reputation score built from completed consults and patient ratings, recorded on-chain.
- Admin verification workflow for new clinician onboarding before marketplace access is granted.

2.3 Follow-Up Engine (Follow-Up Void — 15%) *(Priority 3)*

Runs daily automated post-discharge checkups and escalates anomalies.

- Configurable daily checkup schedule per discharge diagnosis (vitals, symptom questions).
- Delivery via app push, SMS, and USSD depending on patient's registered channel.
- Anomaly-detection rules flag deviating responses for clinician review.
- Missed check-in streaks trigger automatic re-engagement messages and, if unresolved, a clinician alert.
- All checkup responses timestamped and hashed to the patient's on-chain care record.

2.4 Medication Adherence (Medication Misuse — 15%) *(Priority 3)*

Delivers dosage reminders tied to the patient's actual prescription and tracks adherence.

- Prescriptions issued by clinicians are written to an on-chain pharmacy ledger.
- Automated reminders sent at prescribed dosage times via SMS/USSD/push.
- Patient confirms doses taken; missed doses trigger a follow-up nudge and, after repeated misses, a clinician alert.
- Dispensing pharmacies can verify a prescription's authenticity against the ledger before dispensing.

2.5 AI Risk Screening (Late Pathology Discovery — 10%) *(Priority 5)*

Scores symptom, vitals, and basic diagnostic data to flag early-stage risk for referral.

- Risk-scoring model ingests structured symptom intake, vitals trends, and available lab values.
- High-risk flags trigger a routed referral to the nearest facility capable of confirmatory testing.
- Referral outcome fed back into the model training loop for continuous improvement.
- Screening data shared with specialist labs only via on-chain consent tokens.

2.6 Hospital-Blockchain Layer (Cross-Cutting)

Shared trust and audit layer underlying all five modules above.

- Identity & Credentialing: doctor licenses, facility accreditations, and patient identities hashed and anchored on-chain.
- Smart Contract Escrow: consultation fees and gig-doctor payouts held and released on completion conditions.
- Immutable Clinical Ledger: consultations, prescriptions, checkups, and referrals timestamped and hashed on-chain, with encrypted PHI stored off-chain.
- Consent & Data-Sharing Contracts: patient-controlled, revocable access tokens governing record sharing between facilities.

3. Non-Functional Requirements

Category	Requirement
Availability	99.5% uptime target for core API and telemedicine routing during business hours.
Performance	Consult-request routing decision returned in under 5 seconds under normal load.
Offline Access	USSD/SMS channel must function on 2G networks and feature phones with no app install.
Scalability	Horizontally scalable API layer to absorb regional demand spikes (outbreaks, seasonal illness).
Data Protection	PHI encrypted at rest and in transit; only hashes/proofs stored on-chain, never raw PHI.
Compliance	Aligned to applicable national data-protection and health-record regulations per deployment country.
Auditability	Every clinical, payment, and consent action must be traceable to an immutable on-chain record.
Localization	Multi-language support (starting with English and Swahili) across app, SMS, and USSD flows.

4. System Architecture

4.1 Technology Stack

- Web: Next.js 14 (App Router, server components for performance on constrained connections).
- Mobile: React Native with Expo, targeting Android-first given regional device distribution.
- Monorepo pattern with ~70–80% shared business logic across web and mobile via shared packages: ethers.js wrappers, API clients, state management, and data models.
- Backend: Node.js REST API services, containerized for horizontal scaling.
- Database: PostgreSQL with TimescaleDB (vitals/time-series data) and PostGIS (geolocation-based routing).
- Blockchain: EVM-compatible smart contracts for identity, escrow, clinical-record hashing, and consent tokens, accessed via shared ethers.js wrapper package.

4.2 Architecture Diagram

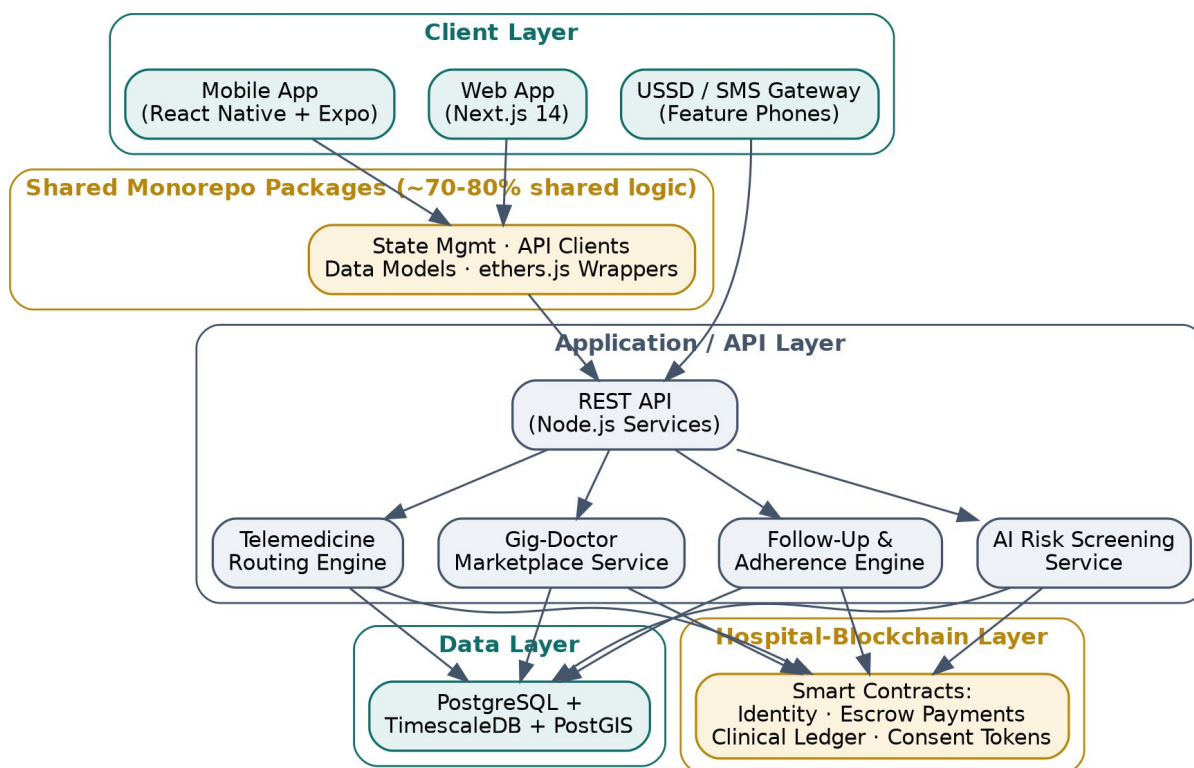


Figure 1. A-health high-level system architecture — client, shared monorepo packages, application services, data layer, and blockchain layer.

5. Database Design (Key Entities)

The schema below highlights the core entities; full column-level detail, indexes, and TimescaleDB hypertable/PostGIS configuration are maintained in the accompanying data dictionary.

5.1 users

Field	Type	Notes
id	UUID (PK)	Shared identity for patients, doctors, admins
role	ENUM	patient doctor admin pharmacy
chain_identity_hash	VARCHAR	On-chain identity anchor reference
phone_number	VARCHAR	Primary channel for SMS/USSD
preferred_language	VARCHAR	en sw ...

5.2 doctors

Field	Type	Notes
id	UUID (PK)	
user_id	UUID (FK → users)	

Field	Type	Notes
credential_hash	VARCHAR	On-chain credential verification reference
specialty	VARCHAR	
verification_status	ENUM	pending verified suspended
reputation_score	DECIMAL	Derived from completed consults/ratings

5.3 consultations

Field	Type	Notes
id	UUID (PK)	
patient_id / doctor_id	UUID (FK)	
channel	ENUM	app sms ussd
status	ENUM	queued in_progress completed escalated
location	GEOGRAPHY (PostGIS)	Used for nearest-clinician routing
ledger_hash	VARCHAR	Anchor to immutable clinical ledger

5.4 checkups

Field	Type	Notes
id	UUID (PK)	
patient_id	UUID (FK)	
recorded_at	TIMESTAMPTZ (Hypertable)	TimescaleDB time-series
vitals / responses	JSONB	Structured intake payload
anomaly_flag	BOOLEAN	
ledger_hash	VARCHAR	

5.5 prescriptions

Field	Type	Notes
id	UUID (PK)	
patient_id / doctor_id	UUID (FK)	
medication / dosage_schedule	JSONB	
adherence_status	ENUM	on_track missed_dose completed
chain_ledger_ref	VARCHAR	Pharmacy ledger anchor

5.6 risk_screenings

Field	Type	Notes
id	UUID (PK)	
patient_id	UUID (FK)	
input_features	JSONB	Symptoms, vitals trend, lab values
risk_score	DECIMAL	
referral_status	ENUM	none referred diagnosed
consent_token_ref	VARCHAR	On-chain consent for data sharing

5.7 payments

Field	Type	Notes
id	UUID (PK)	
payer_id / payee_id	UUID (FK)	
amount / currency	DECIMAL / VARCHAR	
escrow_contract_ref	VARCHAR	Smart contract escrow reference
status	ENUM	held released disputed

6. API Specification (Representative Endpoints)

Method & Path	Description
POST /api/v1/consultations	Create a consult request (app, SMS, or USSD-originated).
GET /api/v1/consultations/:id/status	Poll queue/consult status.
POST /api/v1/doctors/:id/verify	Admin-triggered credential verification, anchors hash on-chain.
POST /api/v1/marketplace/shifts	Doctor accepts an available shift/consult.
POST /api/v1/payments/:id/release	Smart-contract escrow release trigger on service completion.
POST /api/v1/checkups/:patientId	Submit a daily follow-up checkup response.
GET /api/v1/prescriptions/:id/verify	Pharmacy verifies prescription authenticity against ledger.
POST /api/v1/screenings	Submit risk-screening input and receive a risk score.
POST /api/v1/consent/grant	Patient grants/revokes a data-sharing consent token.
POST /ussd/session	USSD gateway session handler for feature-phone flows.

7. Security & Compliance Design

- End-to-end encryption for PHI in transit (TLS 1.3) and at rest (AES-256); only hashes/proofs are written on-chain.
- Role-based access control across patient, doctor, admin, and pharmacy roles, enforced at the API gateway.
- Multi-factor verification for clinician onboarding before marketplace or consult access is granted.

- Consent-token model gives patients granular, revocable control over which facilities/labs can access their records.
- Full audit trail via the immutable clinical ledger, supporting regulatory reporting and dispute resolution.
- Data-residency and national health-data regulation alignment assessed per deployment country prior to go-live.

8. Core Use Cases

Use Case	Primary Actor	Summary
UC1: Patient Consultation	Patient	Initiate and complete a routed telemedicine consult.
UC2: Emergency Transport	Patient / Dispatch	Escalate an urgent finding to rapid-dispatch transport.
UC3: Follow-Up Cycle	Patient / Clinician	Daily post-discharge checkup and anomaly escalation.
UC4: Preventative Screening	Patient / AI Engine	Submit data for AI risk scoring and referral.
UC5: Admin Clinician Verification	Admin	Verify and onboard a new doctor's credentials.
UC6: Gig Shift Fulfillment	Doctor	Accept a shift and receive smart-contract payment.
UC7: USSD Feature-Phone Access	Patient (feature phone)	Access core services without app/internet.

9. 20-Week Implementation Roadmap

The roadmap sequences work by solution weight: the two heaviest-impact gaps (60% combined) anchor Phase 1–3, since a working gig-doctor supply is what lets the telemedicine queue actually clear cases. Follow-Up and Medication modules run in parallel in Phase 4 as they share the same patient-contact and ledger infrastructure. AI Screening is layered on last so it can train against real clinical and adherence data.

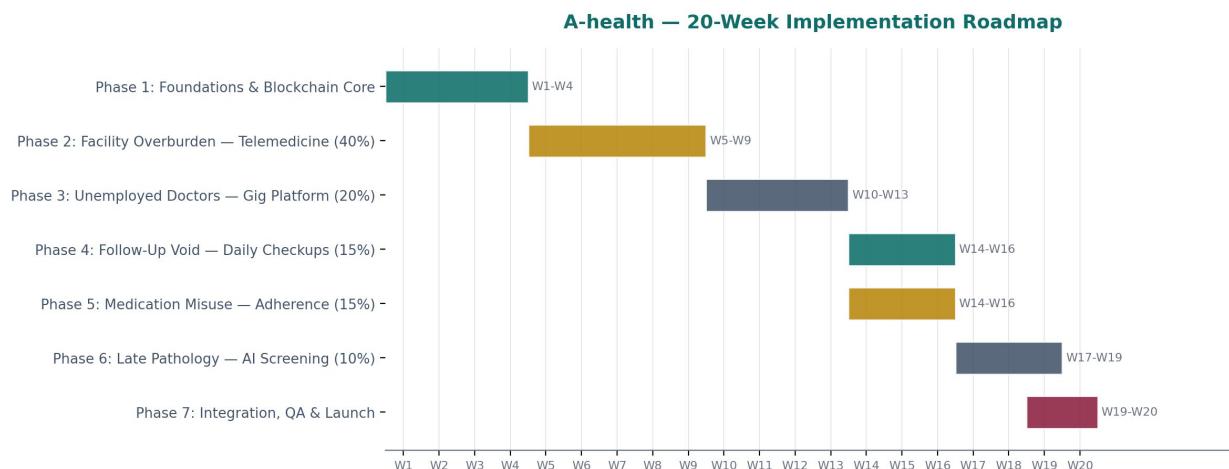


Figure 2. 20-week phase roadmap.

Weeks	Phase	Key Deliverables
1–4	Foundations & Blockchain Core	Monorepo setup, shared packages, DB schema, auth, identity/credentialing smart contracts, CI/CD.
5–9	Facility Overburden — Telemedicine (40%)	Consult intake (app/SMS/USSD), routing engine, queue dashboards, clinical-ledger anchoring.
10–13	Unemployed Doctors — Gig Platform (20%)	Doctor onboarding & verification, marketplace matching, escrow smart contracts, payout flow.
14–16	Follow-Up Void (15%) + Medication Misuse (15%)	Daily checkup engine, anomaly detection, adherence reminders, pharmacy ledger verification.
17–19	Late Pathology Discovery — AI Screening (10%)	Risk-scoring model integration, referral routing, consent-token data sharing.
19–20	Integration, QA & Launch	End-to-end testing, security audit, pilot-facility onboarding, go-live readiness review.

9.1 Week-by-Week Breakdown (Wiki 1 hadi 20)

Jedwali lifuatalo linaonyesha kazi mahususi ya kila wiki, muda (siku za kazi) inayotumika, na moduli inayohusika — ili kuonyesha wazi mtiririko mzima wa wiki 20.

Wiki	Kazi Kuu (Task)	Muda	Moduli
1	Project setup: monorepo (Next.js 14 + React Native/Expo), CI/CD pipeline, dev/staging environments.	5 siku	Foundations
2	Database schema design & migration scripts (PostgreSQL, TimescaleDB, PostGIS); shared data-model package.	5 siku	Foundations
3	Auth & role-based access (patient/doctor/admin/pharmacy); shared API-client package; ethers.js wrapper scaffold.	5 siku	Foundations
4	Identity & credentialing smart contracts (deploy + test on testnet); on-chain identity anchoring service.	5 siku	Blockchain Core
5	Consult-intake UI (app + web) and SMS/USSD intake flow design.	5 siku	Telemedicine (40%)
6	Smart-routing engine v1: nearest-clinician matching using PostGIS + specialty/load rules.	5 siku	Telemedicine (40%)
7	Real-time queue dashboards for patients and clinicians; status polling API.	5 siku	Telemedicine (40%)
8	Emergency-transport escalation workflow + rapid-dispatch integration.	5 siku	Telemedicine (40%)
9	Clinical-ledger anchoring for completed consults; QA & bug-fix pass on telemedicine module.	5 siku	Telemedicine (40%)
10	Doctor registration flow + credential upload; verification queue for admins.	5 siku	Gig Platform (20%)
11	Gig-marketplace matching (shift/demand board by region & specialty).	5 siku	Gig Platform (20%)
12	Escrow smart contracts: payment hold logic + automatic release on completion.	5 siku	Gig Platform (20%)
13	Doctor reputation scoring (on-chain) + payout dashboard; QA pass on gig module.	5 siku	Gig Platform (20%)
14	Daily-checkup engine: scheduling per diagnosis + delivery via app/SMS/USSD.	5 siku	Follow-Up (15%) & Adherence (15%)
15	Anomaly-detection rules for checkup responses; prescription ledger + dosage-reminder engine.	5 siku	Follow-Up (15%) & Adherence (15%)
16	Missed-checkin/missed-dose escalation logic; pharmacy prescription-verification endpoint; QA pass.	5 siku	Follow-Up (15%) & Adherence (15%)
17	Risk-scoring model integration (symptoms, vitals trend, lab values) into API.	5 siku	AI Screening (10%)
18	Referral-routing workflow for high-risk flags; consent-token contracts for data sharing.	5 siku	AI Screening (10%)
19	Model validation against pilot data; start end-to-end integration testing across all modules.	5 siku	AI Screening (10%) + QA
20	Security audit, penetration testing, pilot-facility onboarding, go-live readiness review & launch.	5 siku	Integration, QA & Launch

10. Risks & Assumptions

- Assumes reliable USSD/SMS gateway partnership is secured before Phase 2 begins.
- Assumes pilot facilities can provide at least basic connectivity for clinician-side dashboards.
- Regulatory approval timelines for health-data handling may shift the Phase 6 launch window.
- Doctor supply onboarding (Phase 3) is a dependency for Phase 2 queue-clearing performance and is tracked as a joint risk.